

SC7 Bayes Methods MT25

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Lecture 9 : Approximate Bayesian Computation (ABC)
and simulation-based inference*.

*Methods avoiding likelihood evaluation, just observation model simulations.

ABC - why work hard to fit a model that is wrong anyway?

We use ABC when the likelihood is intractable, but the generative model

$$(\theta, y) \sim \pi(\theta)p(y|\theta)$$

is easy to simulate. Arises in two cases.

(1) If $p(y|\theta) \propto \tilde{p}(y, \theta)$ (as a function of y) then

$$p(y|\theta) = \tilde{p}(y, \theta)/c(\theta)$$

with an intractable normalising constant depending on θ ,

$$c(\theta) = \int \tilde{p}(y, \theta) dy.$$

(2) However, $p(y|\theta)$ may only be *defined by a big simulator*: we don't even have $\tilde{p}(y, \theta)$. This is the main motivation.

These problems are *doubly intractable*. MCMC doesn't work.

ABC example: the Ising model - a doubly intractable model.

Let $\mathcal{Y} = \{0, 1\}^{m^2}$ be all binary images $Y = (Y_1, Y_2, \dots, Y_{m^2})$, $Y_i \in \{0, 1\}$, with cell index $i = 1, \dots, m^2$ on a square lattice.

$\#y$ counts disagreeing neighbors in image $Y = y$.

The *Ising model* is the following distribution over $\mathcal{Y}$:

$$p(y|\theta) = \exp(-\theta \#y) / c(\theta).$$

Here $\theta \geq 0$ is a positive *smoothing parameter* and

$$c(\theta) = \sum_{y \in \mathcal{Y}} \exp(-\theta \#y)$$

is a normalizing constant which we cant compute for m large.*

*There is a formula for $c(\theta)$ for the special case of periodic boundary conditions, but that is not generally our image model of choice.

Observe image data $y_{obs} \in \mathcal{Y}$ and seek estimate of θ .

If $\pi(\theta)$ is a prior for θ then the posterior is

$$\pi(\theta|y_{obs}) = \frac{p(y_{obs}|\theta)\pi(\theta)}{p(y_{obs})}.$$

Consider MCMC targeting $\pi(\theta|y_{obs})$ with a simple θ -proposal, say $\theta' \sim U(\theta - a, \theta + a)$, $a > 0$. The acceptance prob. is

$$\begin{aligned}\alpha(\theta'|\theta) &= \min \left\{ 1, \frac{p(y_{obs}|\theta')\pi(\theta')}{p(y_{obs}|\theta)\pi(\theta)} \right\} \\ &= \min \left\{ 1, \frac{c(\theta)}{c(\theta')} \times e^{(\theta-\theta')\#y} \times \frac{\pi(\theta')}{\pi(\theta)} \right\}\end{aligned}$$

and although $p(y_{obs})$ cancels, $c(\theta)/c(\theta')$ does not, and so we are left with an acceptance probability we cannot evaluate.

Rejection sampling, ABC-style

The basic ABC algorithm approximates the following variant of rejection. Suppose y and θ are both discrete so $p(y|\theta) \leq 1$.

The following simulates $\Theta \sim \pi(\theta|y_{obs})$ where (as usual)

$$\pi(\theta|y_{obs}) \propto p(y_{obs}|\theta)\pi(\theta).$$

(0) Set $n = 0$.

(1) $n \leftarrow n + 1$. Simulate $\theta_n \sim \pi(\cdot)$ and $y_n \sim p(\cdot|\theta)$.

(2) If $y_n = y_{obs}$ return $\Theta = \theta_n$, $N = n$ and stop, else goto (1).

Given θ_n , line (2) succeeds w.p. $\Pr(Y = y_{obs}|\theta_n) = p(y_{obs}|\theta_n)$.

$\Rightarrow$ success at $N = n$ w.p. $\sum_{\theta_n} \pi(\theta_n)p(y_{obs}|\theta_n) = p(y_{obs})$.

Claim: The return value $\Theta \sim \pi(\cdot|y_{obs})$.

Proof: To find the probability $\Theta = \theta$, sum over probabilities for all sequences of rejections ending in the return value θ ,

$$\begin{aligned}\Pr(\Theta_{\text{rej}} = \theta) &= \Pr(\Theta = \theta, N = 1) + \Pr(\Theta = \theta, N = 2) + \dots \\ &= \pi(\theta)p(y_{obs}|\theta)(1 + r + r^2 + \dots) \\ &= \pi(\theta)p(y_{obs}|\theta) \times \frac{1}{1 - (1 - p(y_{obs}))} \\ &= \pi(\theta|y_{obs})\end{aligned}$$

since we reject at each pass over (1)-(2) w.p. $r = 1 - p(y_{obs})$.

Rejection-ABC (where now $(\Theta, Y) \in \Omega \times \mathcal{Y}$ - general rv).

If y is “close” to y_{obs} then $\pi(\theta|y)$ approximates $\pi(\theta|y_{obs})$.

Measure “close” using summary statistics $S(y) = (S_1(y), \dots, S_p(y))$ chosen to inform θ .

Let $D(s, s_{obs})$ be distance between $s = S(y)$, $s_{obs} = S(y_{obs})$.

Let $\delta \geq 0$ be a threshold distance.

(0) Set $n = 0$.

(1) $n \leftarrow n + 1$. Simulate $\theta_n \sim \pi(\cdot)$ and $y_n \sim p(\cdot|\theta)$.

(2) If $D(S(y), s_{obs}) < \delta$ stop with $\Theta = \theta_n$, $N = n$, else goto (1).

Claim: Define $\pi_{ABC}(\theta|y_{obs}) = \pi(\theta|Y \in \Delta_\delta(y_{obs}))$, $\theta \in \Omega$.

The distribution of the return value is $\Theta \sim \pi_{ABC}(\cdot|y_{obs})$.

Proof: Take $A \in \mathcal{B}_\Omega$ and show $\Pr(\Theta \in A) = \pi_{ABC}(A|y_{obs})$.

Let $\Delta_\delta(y_{obs}) = \{y \in \mathcal{Y} : D(S(y), s_{obs}) < \delta\}$.

$\{\Theta \in A, N = n\} = \{y_{<n} \notin \Delta_\delta(y_{obs}), (\theta_n, y_n) \in A \times \Delta_\delta(y_{obs})\}$.

Accept at (1)-(2) w.p. $p(\Delta_\delta(y_{obs})) = \int_{\Delta_\delta(y_{obs})} \int_\Omega \pi(\theta) p(y|\theta) d\theta dy$.

$$\begin{aligned} \Pr(\Theta \in A) &= \sum_{n=1}^{\infty} \Pr(\Theta \in A, N = n) \\ &= \sum_{n=1}^{\infty} \Pr((\theta_n, y_n) \in A \times \Delta_\delta(y_{obs}), y_{<n} \notin \Delta_\delta(y_{obs})) \\ &= \sum_{n=1}^{\infty} \Pr((\theta_n, y_n) \in A \times \Delta_\delta(y_{obs})) (1 - p(\Delta_\delta(y_{obs})))^{n-1} \\ &= \frac{\int_A \int_{\Delta_\delta(y_{obs})} p(y|\theta) dy \pi(\theta) d\theta}{p(\Delta_\delta(y_{obs}))} = \int_A \frac{p(\Delta_\delta(y_{obs})|\theta) \pi(\theta) d\theta}{p(\Delta_\delta(y_{obs}))} \\ &= \pi_{ABC}(A|y) \end{aligned}$$

We have made two (more) approximations

- we summarise the data with $y_{obs} \rightarrow s(y_{obs})$
- we replace the data-statement

“the realised value of the data is $Y = y_{obs}$ ”

with

“the realised value of the data is in the ball $Y \in \Delta_\delta(y_{obs})$ ”.

If $S : \mathcal{Y} \rightarrow R^p$ is a sufficient statistic then

$$\pi(\theta|Y = y_{obs}) = \pi(\theta|S(Y) = s(y_{obs})),$$

and in that case

$$\pi(\theta|Y \in \Delta_\delta(y_{obs})) \rightarrow \pi(\theta|y_{obs})$$

as $\delta \rightarrow 0$ may often be verified.

Example

Data model: $y_i \sim \text{Poisson}(\Lambda)^*$, $i = 1, 2, \dots, n$ with $n = 5$.

Prior: $\Lambda \sim \Gamma(\alpha = 1, \beta = 1)$.

Summary statistic: $S(y) = \bar{y}$

Distance measure: $D(\bar{y}, \bar{y}_{obs}) = |\bar{y} - \bar{y}_{obs}|$ and $\delta = 0.5, 1$.

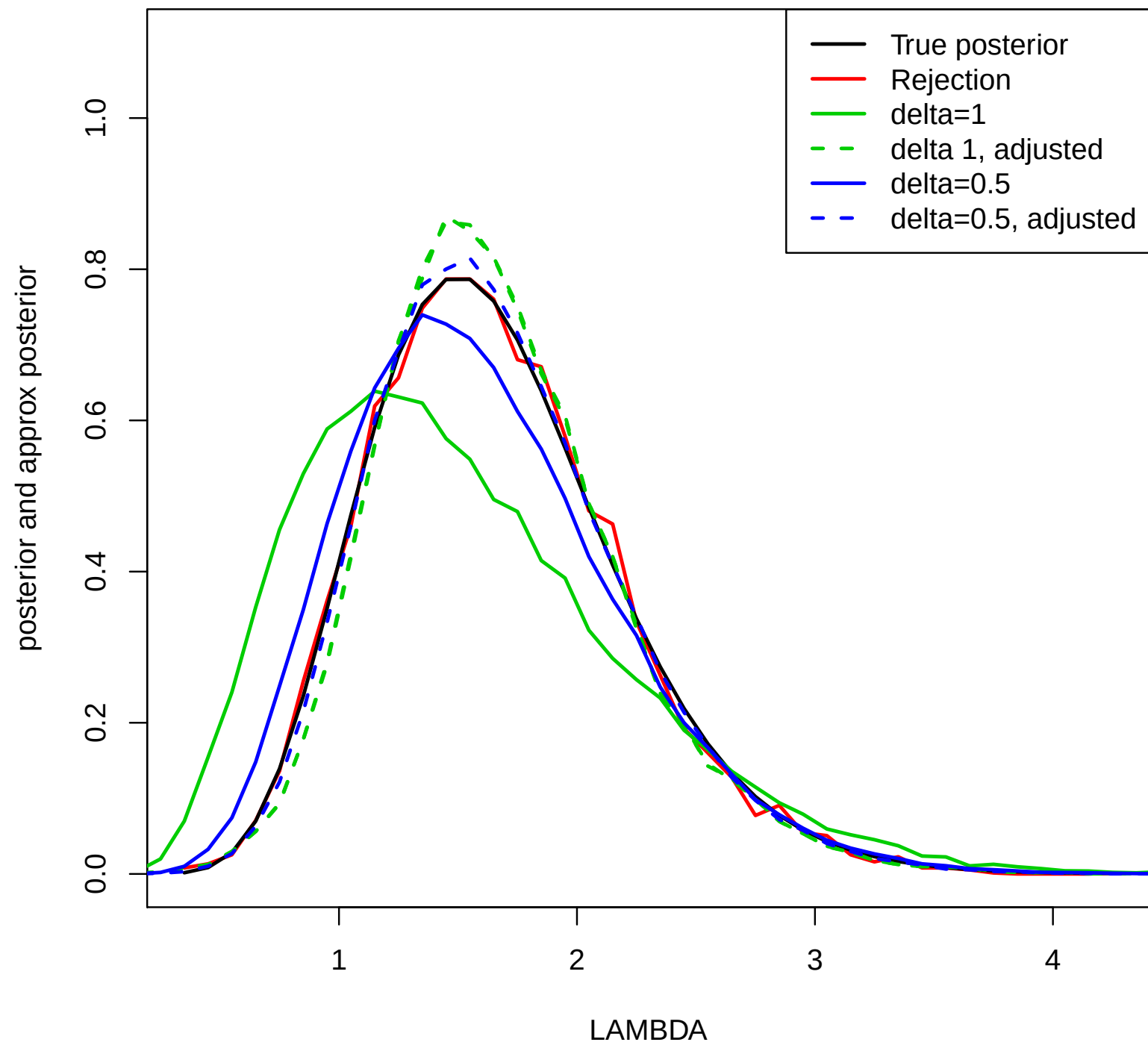
ABC algorithm

(1) Simulate $\lambda \sim \Gamma(\alpha, \beta)$ and $y_i \sim \text{Poisson}(\lambda)$, $i = 1, 2, \dots, n$.

(2) If $|\bar{y} - \bar{y}_{obs}| < \delta$ return λ and stop, otherwise goto (1).

We do Bayesian inference without calculating $p(y_{obs}|\theta)$ or $\pi(\theta)$, sometimes called “likelihood-free” or “simulation-based” inference. We just specify how to simulate parameters and data.

*True value was $\Lambda = 2$.



Regression adjustment of samples

ABC generates $\theta, y \sim \pi(\theta)p(y|\theta)$. Conditional on y , $\theta \sim \pi(\theta|y)$.
“Shift” this distribution to match $\pi(\theta|y_{obs})$.

Let $\theta \sim \pi(\cdot|y)$, $\theta' \sim \pi(\cdot|y_{obs})$, $s = S(y)$ and $s_{obs} = S(y_{obs})$.

Assume (1) The posterior mean $\mu(S(y)) = E(\theta|Y = y)$ is a linear function of $S(y)$ alone, that is, for some $\beta \in R^p$,

$$\mu(s) = \mu(s_{obs}) + (s - s_{obs})^T \beta.$$

Assume (2) The random variables $\theta - \mu(s)$ and $\theta' - \mu(s_{obs})$ are identically distributed.

A1 is good at small $|s - s_{obs}|$ (Taylor series) when s sufficient.

Claim: Let $\theta_{adj} = \theta - (s - s_{obs})\beta$. If A1 & A2 then

$$\theta_{adj} \sim \pi(\cdot | y_{obs})$$

Proof: By A2 $\theta \sim \theta' + \mu(s) - \mu(s_{obs})$, so

$$\theta - (s - s_{obs})^T \beta \sim \theta' + \mu(s) - \mu(s_{obs}) - (s - s_{obs})^T \beta$$

and so by A1 $\theta - (s - s_{obs})^T \beta \sim \theta'$.

Claim: Let $\theta_{adj} = \theta - (s - s_{obs})\beta$. If A1 & A2 then

$$\theta_{adj} \sim \pi(\cdot | y_{obs})$$

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and so by A1 $\theta - (s - s_{obs})^T \beta \sim \theta'$.

Regression adjustment Let $\epsilon \sim \theta - \mu(s)$ so $\theta \sim \mu(s) + \epsilon$ and

$$\theta \sim \mu(s_{obs}) + (s - s_{obs})^T \beta + \epsilon \quad \text{by A1.}$$

Estimate β by regressing θ on $(s - s_{obs})$:

(1) run ABC at fixed δ and get pairs of samples $(\theta^{(t)}, y^{(t)})_{t=1}^T$.

(2) Let $s^{(t)} = S(y^{(t)})$. Take

$$(\hat{\alpha}, \hat{\beta}) = \arg \min_{\alpha, \beta} \sum_{t=1}^T (\alpha + (s^{(t)} - s_{obs})^T \beta - \theta^{(t)})^2$$

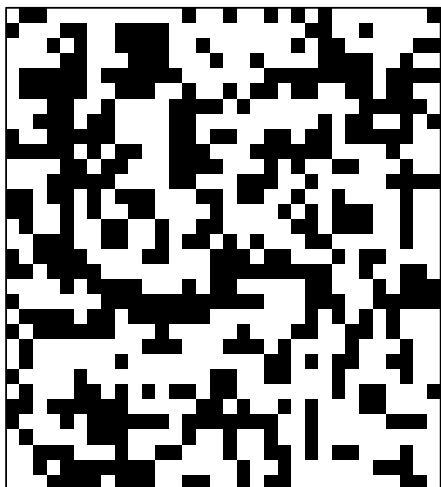
If $\hat{\theta}_{adj}^{(t)} = \theta^{(t)} - (s^{(t)} - s_{obs})^T \hat{\beta}$ then

$$\hat{\theta}_{adj}^{(t)} \sim \pi(\cdot | y_{obs}) \quad (\text{approx}).$$

ABC example: the Ising Model

The Ising distribution $Y \sim p(\cdot|\theta)$ is easy to sample for moderate m -values using MCMC. Here are 3 samples $Y \sim p(\cdot|\theta)$:

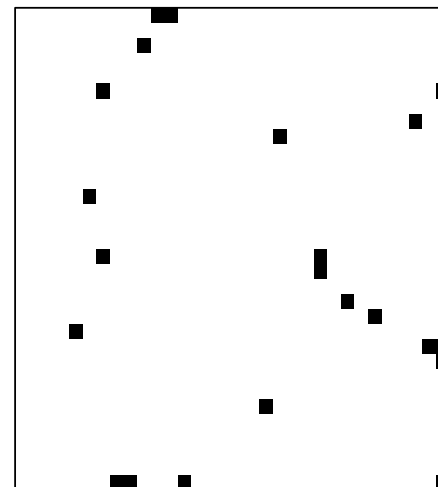
Ising model, theta=0.4



Ising model, theta=0.8



Ising model, theta=1.2



These samples are not exactly distributed according to $p(y_{obs}|\theta)$ (convergence) but we can make them as good as we need by taking long MCMC runs.

ABC inference for θ for Ising model

Recall the doubly intractable inference for $\theta|y_{obs}$.

We have data $Y = y$ which is an $m \times m$ binary matrix and our observation model for Y , $Y \sim p(y|\theta)$ is the Ising model. Our goal is to estimate θ . The posterior is

$$\pi(\theta|y_{obs}) = \frac{p(y_{obs}|\theta)\pi(\theta)}{p(y_{obs})}$$

and the likelihood

$$p(y_{obs}|\theta) = \exp(-\theta \# y_{obs}) / c(\theta)$$

depends on $c(\theta)$, an intractable function of θ .

ABC algorithm sampling $\theta \sim \pi(\theta|y_{obs})$ approximately:

Prior $\theta \sim \text{Exp}(2)$ has right scale.

Statistic: $S(y_{obs}) = \#y_{obs}$, (which is sufficient).

(1) Simulate $\theta \sim \text{Exp}(2)$ and $y \sim \exp(-\theta \#y)/c(\theta)$ (MCMC).

(2) If $|\#y - \#y_{obs}| < \delta \times 2m(m-1)^*$ return θ , else goto (1).

Implemented in online code: 8x8 Ising, true $\theta = 0.8$.

The ABC-posterior converges to true posterior as $\delta \rightarrow 0$.

The regression adjustment for $\delta = 0.1$ corrects its distribution to agree with that for $\delta = 0.05$.

*scale by $\max(\#y)$

